



ISTANBUL TECHNICAL UNIVERSITY

Department of Electrical Engineering

ENGINEERING DRAWING

(ELK 363E)

Pictorial Sketching

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10/30/2023

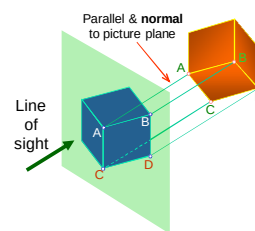
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Objectives

- Be able to **explain the difference** between an **axonometric projection** and an **oblique projection**.
- Be able to **explain the difference** between an **isometric projection** and an **isometric drawing/sketch**.
- Be able to **create** an **isometric** and **oblique sketches** from an actual object and multiview drawing.

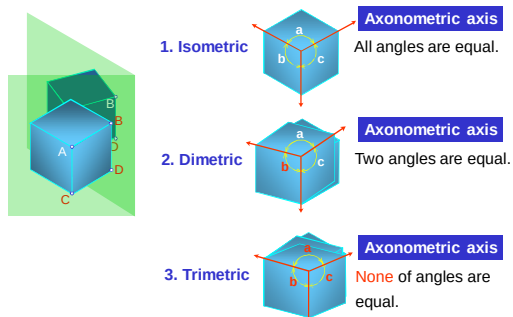
Axonometric & Oblique Projection

Axonometric Projection

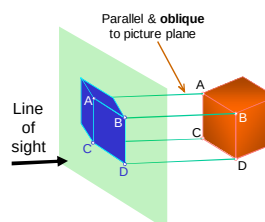


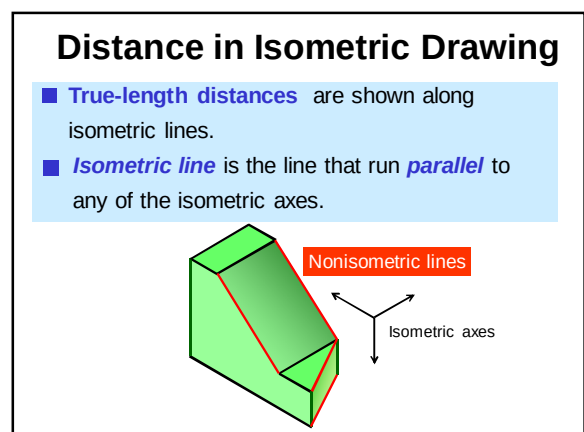
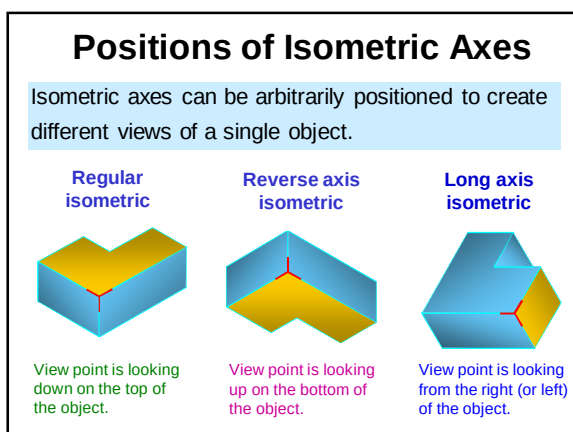
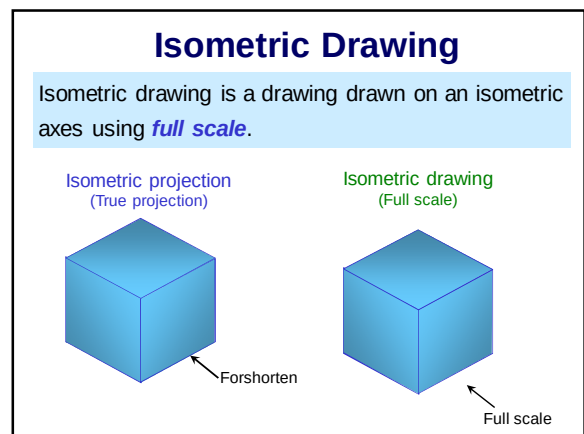
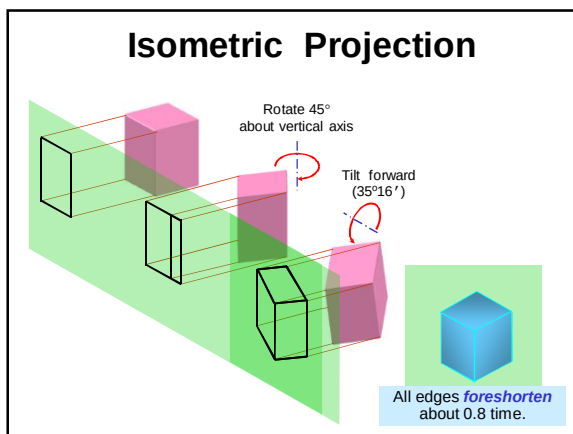
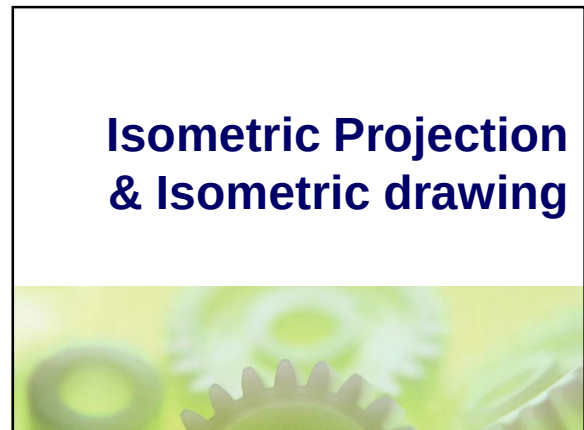
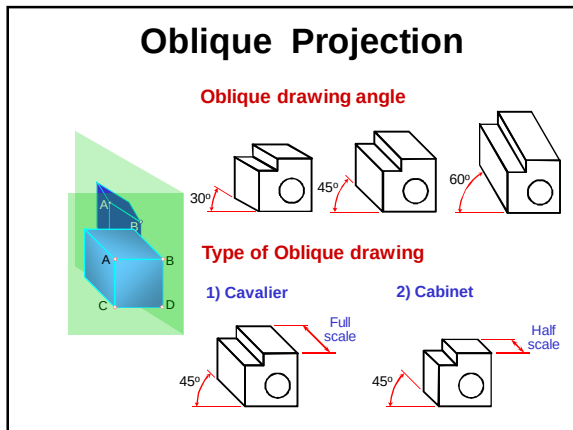
Axonometric Projection

Type of axonometric drawing



Oblique Projection





Isometric Sketching



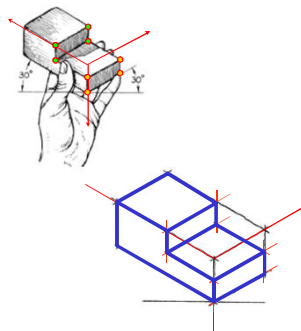
Sketch from an actual object

1. Place the object in the position which its shape and features are clearly seen.
2. Define an isometric axis.
3. Sketching the enclosing box.
4. Estimate the size and relationship of each details.
5. Darken all visible lines.

Sketch from an actual object

STEPS

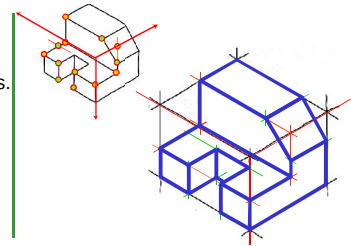
1. Positioning object.
2. Select isometric axis.
3. Sketch enclosing box.
4. Add details.
5. Darken visible lines.



Sketch from an actual object

STEPS

1. Positioning object.
2. Select isometric axis.
3. Sketch enclosing box.
4. Add details.
5. Darken visible lines.

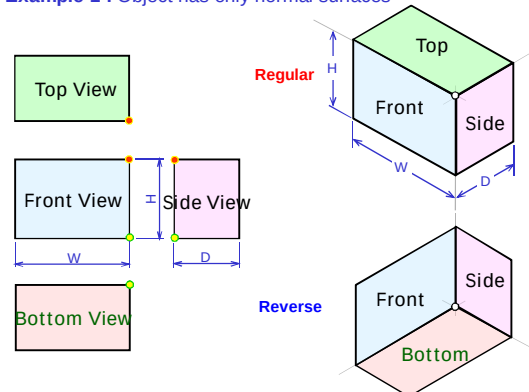


Note In isometric sketch/drawing, hidden lines are **omitted** unless they are absolutely necessary to completely describe the object.

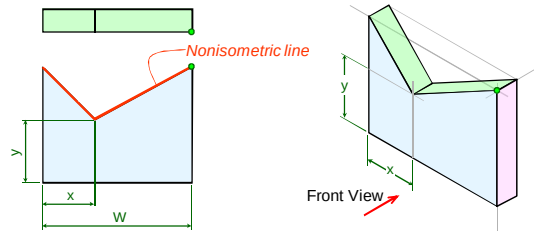
Sketch from multiview drawing

1. Interpret the **meaning of lines/areas** in multiview drawing.
2. Locate the lines or surfaces relative to isometric axis.

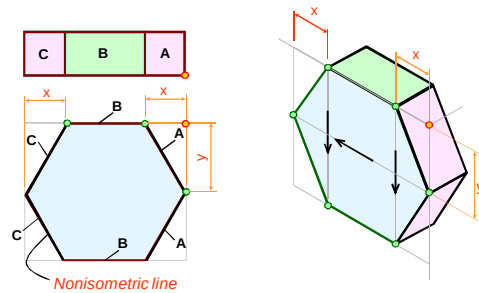
Example 1 : Object has only normal surfaces



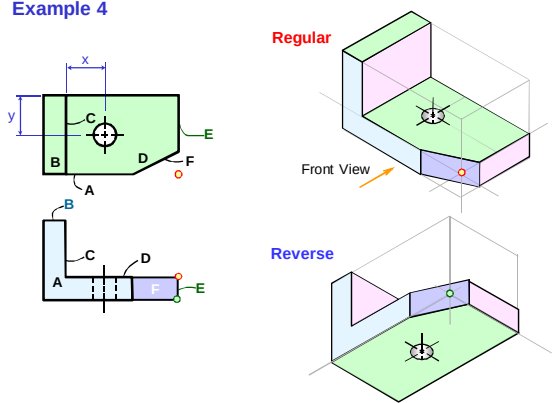
Example 2 : Object has inclined surfaces



Example 3 : Object has inclined surfaces



Example 4

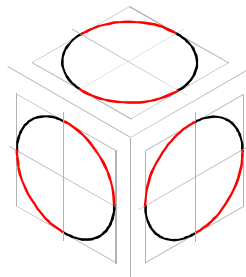


Circle & Arc in Isometric

■ In isometric drawing, a circle appears as an ellipse.

Sketching Steps

1. Locate the center of an ellipse.
2. Construct an isometric square.
3. Sketch arcs that connect the tangent points.

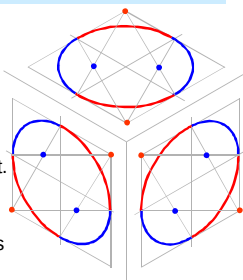


Circle & Arc in Isometric

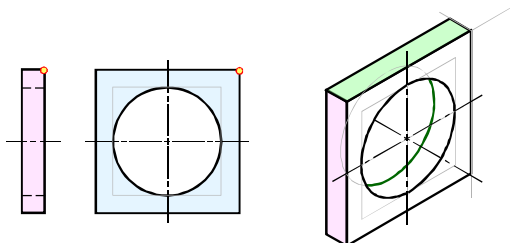
■ **Four-center** method is usually used when drawn an isometric ellipse with drawing instrument.

Sketching Steps

1. Locate the center of an ellipse.
2. Construct an isometric square.
3. Construct a perpendicular bisector from each tangent point.
4. Locate the **four** centers.
5. Draw the arcs with these centers and tangent to isometric square.



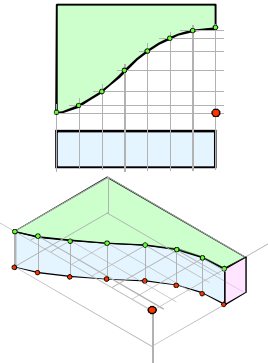
Example 5



Irregular Curve in Isometric

Steps

1. Construct points along the curve in multiview drawing.
2. Locate these points in the isometric view.
3. Sketch the connecting lines.

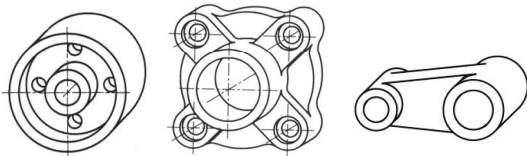


Oblique Sketching



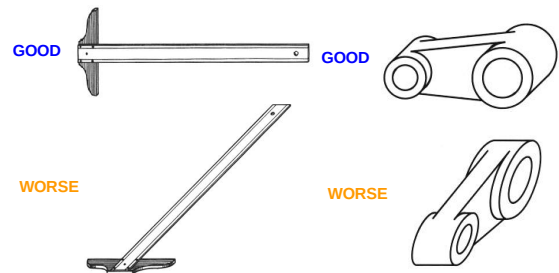
Object Orientation Guidelines

- Place complex features (arc, hole, irregular shape surface) parallel to frontal plane.



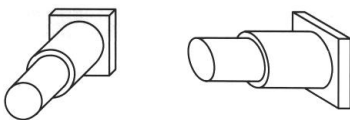
Object Orientation Guidelines

- The longest dimension of an object should be parallel to the frontal plane.

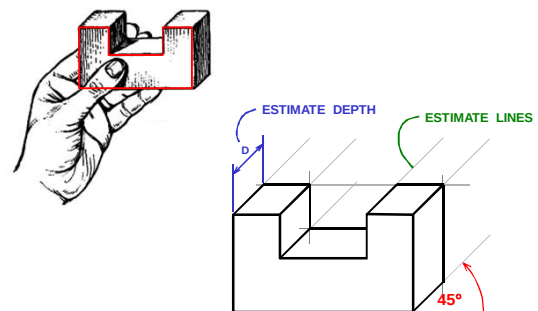


Object Orientation Guidelines

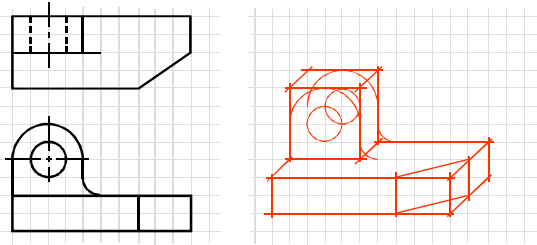
Which orientation is better ?



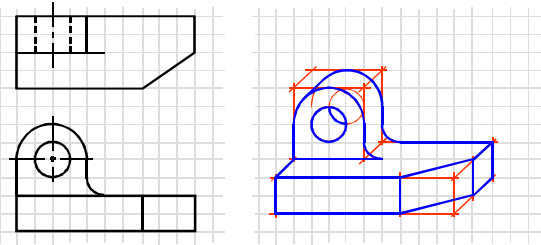
Sketch from actual object



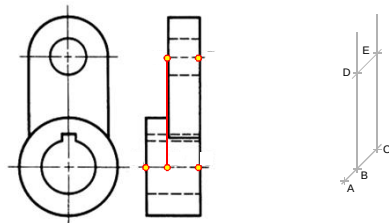
Sketch from multiview drawing



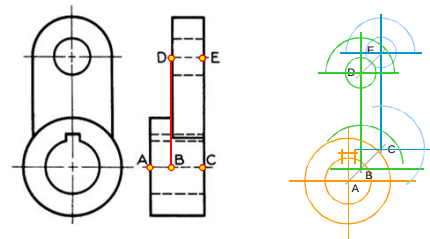
Sketch from multiview drawing



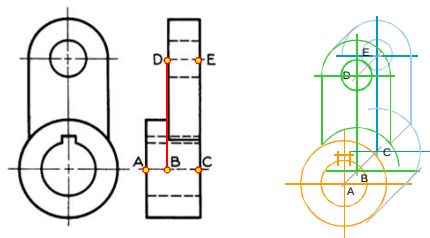
Sketch from multiview drawing



Sketch from multiview drawing



Sketch from multiview drawing



Sketch from multiview drawing

